



SMARTRENT AI

DECISION INTELLIGENCE FOR URBAN HOUSING

Empowering Renters with AI-Driven Economic Clarity

Hackathon
Project

Pitch Deck





THE PROBLEM

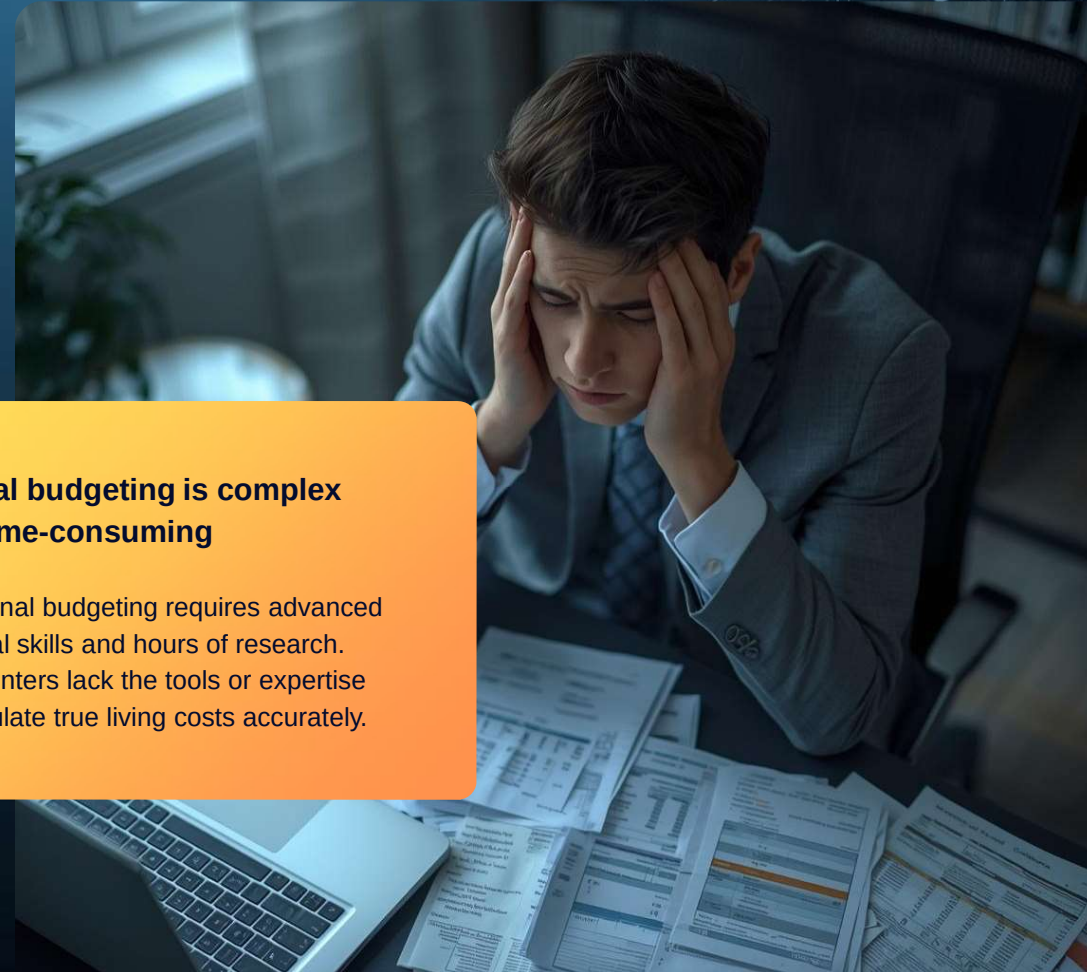
ECONOMIC BLINDNESS

Renters focus only on monthly rent, ignoring hidden costs

Transport, utilities, maintenance, and lifestyle expenses are overlooked. This economic blindness leads to financial instability and poor housing choices.

Manual budgeting is complex and time-consuming

Traditional budgeting requires advanced financial skills and hours of research. Most renters lack the tools or expertise to calculate true living costs accurately.



AI-Powered Integration

Our AI engine seamlessly connects salary inputs, financial commitments, transport logistics, and rental advertisements into one unified analysis framework.

Curator Role Shift

Transforms users from tedious manual researchers into empowered financial curators who simply review AI-generated insights.

Real Cost of Living Mock-up

Instantly generates high-fidelity economic mock-ups showing true housing costs beyond just monthly rent—including all hidden expenses.

Instant Decision-Making

Accelerates rental decision preparation from hours of spreadsheet work to mere seconds of intelligent analysis.

OUR SOLUTION

SmartRent AI is an AI-native financial curation engine that transforms how renters evaluate housing options. By integrating salary data, existing commitments, transport costs, and rental listings, it instantly generates a comprehensive "Real Cost of Living" economic analysis—shifting users from manual researchers to empowered financial curators.





KEY FEATURES

SmartRent AI delivers powerful capabilities that transform how renters evaluate housing options. Our AI engine parses unstructured rental critical data, and provides instant financial clarity. Compare multiple options simultaneously on an intuitive 2D canvas with real-time cost analysis.

Track unlimited financial scenarios with version control. Edit variables using natural language commands for seamless budget adjustments.



SmartRent AI
DECISION INTELLIGENCE ENGINE PENGUN_LIMHACKTHON_2028

FINANCIAL PROFILE

MONTHLY SALARY (RM)
4500

FIXED COMMITMENTS
1800

MAX DEPOSIT BUDGET
1494

COMMUTE LOGISTICS

WORKPLACE (AREA)
sungai besi

TRANSPORT MODE
Public Transport

MAX DISTANCE (KM)
16

RENTAL ADS (MAX 5)

+ Add Option

#01 <https://www.mudah.my/2-storey-townhouse-at-taman-sungai-bi>

PENGUN

Data Parsing AI

Multi-Option Comparison

Financial Versioning



Vercel Serverless Functions

Frontend: Responsive HTML5/JS with Tailwind CSS powers the dynamic comparison canvas.

Backend: Stateless Vercel Serverless Functions provide instant scalability and zero cold-start delays for real-time user interactions.



Z.AI GLM central reasoning engine



Token Management for efficiency



Context Construction for accurate reasoning

SYSTEM ARCHITECTURE

Our architecture leverages Vercel Serverless Functions for stateless, scalable backend operations. Requests are relayed to the Z.AI GLM API, which serves as the central reasoning engine performing multi-step financial logic and real cost calculations.

Token Management and Context Construction modules ensure efficient API usage and accurate financial reasoning outputs.

PROTOTYPE DEMO OVERVIEW

> Live dashboard showcasing AI in action

Our working prototype demonstrates the full SmartRent AI workflow: users input salary and monthly commitments and select commute options. The system instantly parses unstructured data and calculates Real Cost of Living with automatic safety buffer enforcement.



smart-rent-five.vercel.app

MAX DISTANCE
20 KM

PASSED

#AD2 Sungai Besi

~7 KM commute

Rental Price:
Logistics Cost:

REAL COST (COL):

Z.AI: Optimal match found ba

57

LEFTOVER CASH

RM 4087

Update profile? (e.g. Salary 5000)



Multi-Option Comparison

Compare up to 5 rental options side-by-side with leftover cash values for Cheras, Sungai Besi, Bangsar and more locations.



Real Cost Indicators

PASSED/CAUTION status flags instantly show which properties meet the RM 550 safety buffer threshold for sustainable living.

 FINANCIAL PROFILE

MONTHLY SALARY (RM)

5

FIXED COMMITMENTS

e.g. 1200

MAX DEPOSIT BUDGET

e.g. 3500



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RM 550 Safety Buffer

Mandatory disposable income threshold embedded in all calculations ensures renters maintain financial sustainability after housing costs.

100% Jest Test Pass Rate

Comprehensive Jest framework testing validates all safety buffer calculations with 100% pass rate, ensuring reliable financial guidance.

Intelligent Risk Flagging

System automatically flags at-risk properties where disposable income falls below the safety threshold, alerting users to potential financial strain.

Graceful Error Handling

Robust retry paths and adversarial prompt prevention maintain system integrity and output accuracy even under edge cases.

SAFETY BUFFERS

SmartRent AI prioritizes financial safety through embedded safeguards. Our mandatory RM 550 disposable income buffer ensures renters maintain sustainable finances. Combined with rigorous testing and error handling, we deliver trustworthy recommendations that protect users from risky housing decisions.





Persistent sessions & GPS integration

Our roadmap prioritizes scalable infrastructure and user-centric features. Advanced capabilities including PDF reporting for exportable financial summaries, multi-user collaboration for shared household decisions, and A/B testing frameworks will enable continuous optimization and feature refinement.



PDF reporting & export features



Multi-user collaboration tools



A/B testing for continuous optimization

TECHNICAL SCALABILITY

Current MVP delivers stateless, fast API architecture with manual commute inputs. Future development includes persistent user sessions for saved preferences, real-time GPS integration for automatic commute calculation, and robust database support for historical data tracking.

Gradual rollout via grayscale release ensures stability and user feedback integration before full deployment.

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THANK YOU FOR YOUR ATTENTION

SmartRent AI transforms rental decisions from guesswork to data-driven confidence.
Join us in revolutionizing urban housing decisions.



<https://smart-rent-five.vercel.app/>